

Braillix - Electronics BOM (HAVE on top, BUY at bottom)

Written 2026-07-30. Companion to docs/MASTER_BOM.md (which also covers mechanical parts). Scope: **one working single-cell demo**. Multi-cell/production parts are listed separately at the end so they are not confused with what you need now.

The headline: you need NO chips, NO resistors, NO capacitors

Every discrete component the circuit would normally need is **already built into the modules you own**:

What you'd normally add	Why you don't need it
Motor flyback diodes	Built into the ULN2003A. The chip has internal freewheeling diodes on its COM pin - that is what COM is for.
Motor driver transistors	That IS the ULN2003. Seven Darlington pairs, 500mA each. You need four.
Button pull-up resistors	ESP32 has internal pull-ups. Firmware uses INPUT_PULLUP.
Hall sensor pull-up / comparator	On the blue MH module already (that's the second IC and the trimmer pot).
Voltage regulator + decoupling caps	On the ESP32 DevKit already (AMS1117 + caps).
Reverse-polarity protection	[!] NOT present in the circuit. Mitigated by measurement: the owned jack was tested and is correct. Measure any new connector before trusting it.

Total passives to buy for the demo: zero.

PART 1 - WHAT YOU ALREADY HAVE

Cross-check each of these physically. The "how to identify it" column is so you can confirm you actually have the right thing.

#	Component	How to identify it	Qty	Status
1	ESP32 DevKit (USB-C)	Black PCB ~51x28mm, silver square shield, USB-C at one end, 15 pins per side, BOOT + EN buttons	1	[OK] HAVE (?350)
2	28BYJ-48 stepper motor	Silver can ~28mm dia, blue plastic base, 5 coloured wires into a white plug	1	[OK] HAVE (?160 w/ driver)
3	ULN2003 driver board	Small blue PCB, 4 red LEDs, white 5-pin socket, black 16-pin chip, IN1-IN4 + +/- pins	1	[OK] HAVE - this is your motor driver, no chip to buy
4	Hall sensor module (MH-Sensor-Series)	Small blue PCB, tiny black 3-pin sensor on one edge, blue trimmer pot, 1-2 LEDs, pins marked AO DO GND VCC	1	[OK] HAVE (?50) - [!] must use AO (analog) , and the pocket won't fit it (see note)
5	5V / 3A power adapter	Wall plug, barrel connector on the lead, "5V 3A" on the label	1	[OK] HAVE (?160 w/ jack)
6	Inline female DC pigtail jack	Black cylindrical barrel socket with red/black wires; no thread or nut	1	[OK] HAVE / CURRENT POD CHOICE; body stays outside, leads pass through the selected lid slot, internal tie post carries pull load. [OK] Polarity measured correct 2026-08-01 (red = +).

#	Component	How to identify it	Qty	Status
6b	5.5x2.1mm female panel-mount jack	Threaded neck plus retaining nut	0	? NOT REQUIRED for the current inline-pigtail revision
7	Connecting wire (2m)	Loose hookup wire	2m	[OK] HAVE (?20)
8	Dupont jumper wires	Ribbon of coloured wires with plastic pin housings	some	[OK] HAVE - need ~8x M-F and ~5x M-M
9	Neodymium magnets 8x1mm	Small silver discs, strongly attract each other	10	[OK] HAVE (?135) - for docking, not for cam homing
10	Tactile switches 6x6x5mm	Tiny square black buttons, 4 legs	3	[OK] HAVE - not needed for the demo
11	Digital multimeter	Handheld meter with DC volts	1	[OK] HAVE (bought 2026-08-01) - jack polarity measured and CONFIRMED correct

Notes on things you already have

#6 - [OK] polarity is VERIFIED CORRECT (2026-08-01). Measured with the multimeter: **red = positive**. This retires the single largest risk to the ESP32. Mark the positive wire physically (nail polish, marker, tag) so the information cannot be lost at the bench. There is still no reverse-polarity protection in the circuit, so if this connector is ever replaced, the replacement must be measured the same way before it is trusted.

#4 - your whole Hall module does not fit the base plate. v7.5 rebuilt an underside pocket for the *bare* TO-92 sensor only; the blue PCB is still far too large. M11b measures 1.6mm and the current recess is exactly 1.6mm, so dry-fit it before gluing. **For the demo this does not matter** - the module sits on the breadboard. For final assembly, desolder the bare 3-pin sensor from the module and run three wires back to it.

PART 2 - WHAT YOU NEED TO BUY

2A. For the demo (buy this week)

#	Component	Spec	Why	Est. ?
1	Soldering station, 60W, TEMPERATURE-CONTROLLED	"936" type, ceramic heater, 2.4mm chisel tip	[!!] NOT OWNED. [!] Do NOT buy a plain ?200 pencil iron - uncontrolled heat burns the flux off and makes soldering feel impossible. See ELECTRONICS_PLAN.md Part 9.	~1,300
2	Breadboard	Half-size or full-size, 400-830 tie points	The whole Tier-0 demo is breadboard-based, and it's the safe way to bring up wiring before soldering.	~100
3	USB-C cable (DATA)	Must carry data, not charge-only	To flash the ESP32. Charge-only cables look identical and fail silently. Test the one you have first - you may already be fine.	~150
4	Solder wire	0.8mm, 60/40 rosin core	For soldering wires flat to the ULN2003. Not plumbing solder.	~150
5	Wire stripper / cutter	Small, for 22-26 AWG	Prepping wire ends.	~200

Demo subtotal: ~?1,900 (or ~?1,750 if your existing USB-C cable does data). The soldering station is the bulk of it and is the one genuinely missing tool.

2B. For the mechanism (buy when the resin parts arrive)

#	Component	Spec	Qty	Est. ?
6	Micro compression springs	2.0mm OD, ~0.3mm stainless wire, ~4mm free length. Buy an assortment kit. [!!] Pen springs (4mm) cannot work - braille rows are 2.6mm apart.	6 + spares	~500
7	Homing magnet	[!!] 3mm dia x 1mm thick neodymium - CHANGED from 3x2mm in v7.5 . The cam disc floor is only 2mm, so a 2mm magnet's pocket cut clean through it and punched a hole across three cam tracks. 1mm leaves 0.8mm of floor and couples just as well. Your 8x1mm ones are too wide.	1 (buy 10)	~120
8	M2.5 x 25mm bolts	Top plate -> standoffs -> box bosses	4	~60
9	M4 x 5mm or 6mm thread-forming screws	Candidate range for the approved raised-stack design. Select the pilot with base_interface_coupon; measure the real ear and screw point; require >=0.5mm clearance below the cam. Never use M4x10 . No nuts.	2	~30
10	M2 x 8mm self-tap	Pod lid. NOT M2x6 - through a 4mm lid that leaves only 2mm of thread.	2	~40
11	Superglue or 5-min epoxy	Glues the resin dot insert into the PETG plate. Epoxy preferred - you get time to seat it square.	1	~80
12	Digital calipers	150mm digital	Already in hand; M5, M11b, and M21 recorded on 2026-07-31.	1

Mechanism subtotal: ~\$1,430

2C. Optional / nice to have

Component	Why	Est. ?
Hot glue gun + sticks	* Strain relief on every joint. Reworkable - this is why it beats epoxy while the pin map can still change. See ELECTRONICS_PLAN.md Part 12.	~250
Neutral-cure RTV silicone	Optional upgrade over hot glue for vibration. [!!] Must say neutral cure - the vinegar-smelling acetic kind corrodes copper.	~150
Flux paste	Makes soldering to ULN2003 pins much easier for a beginner	~100
Helping-hands / third hand	Not really optional - you cannot hold iron, solder and two wires with two hands	~200
Brass wool tip cleaner	Better than the wet sponge; keeps the tip alive	~100
Heat-shrink tubing	Insulating soldered joints	~80
Tweezers	Handling 2mm springs and 1mm linkages - genuinely hard without	~80

Component	Why	Est. ?
Desoldering pump/wick	Fixing bridges; also for lifting the TO-92 off the hall module	~120
Small drill bits 1.5/1.7/2.5mm	FDM holes print 0.2-0.3mm undersize	~150
Electrical tape	-	~30

2D. Cost of each ADDITIONAL cell (the scaling claim, priced)

You are building two. This is what a third, fourth or eighth costs - no custom parts at any point. Full architecture in ELECTRONICS_PLAN.md Part 5.

Per extra cell	Est. ?
28BYJ-48 motor + ULN2003 board	~160
Hall sensor module	~50
MCP23017 expander (DIP-28) + 3 address jumpers	~90
Wire, connector, heat-shrink	~50
Total per cell	~?350

One-off, only when you go past 2 cells: 2x 4.7k Ω pull-ups (~?5, brain end only). **One-off, only past 16 cells:** TCA9548A I2C mux (~?120).

[!] Cells 1 and 2 run **direct-GPIO** and need no expander at all - but building both muscle boards with the expander anyway is what makes the N-cell claim demonstrable rather than theoretical.

Totals

	?
Already spent (incl. multimeter)	~1,435
Demo essentials (2A)	~1,900
Mechanism (2B)	~1,430
Optional (2C)	~960
Everything	~5,725 of ?15,000
each cell beyond the 2nd (2D)	*~350*

The breadboard demo needs NO soldering at all - only the breadboard and a data cable (~?250). The soldering station is required to *assemble* a cell, not to demonstrate one.

PART 3 - NOT NEEDED (so you don't buy it by mistake)

Item	Verdict
Any motor driver IC	[X] You own it. The chip on the blue board is the ULN2003A.
ATmega328P / custom muscle board	[X] Never. Multi-cell is solved by an MCP23017 per brick (DIP-28, hand-solderable, ~?80) - see ELECTRONICS_PLAN.md Part 5. TQFP-32 at 0.8mm pitch is beyond basic soldering and would need PCBA ordering.
Resistors	[X] None for one or two cells. *(From 3 cells: exactly 2x 4.7k Ω I2C pull-ups, at the brain end only, once, ever - repeating them per brick kills the bus.)*
Capacitors	[X] None. Both modules and the DevKit carry their own.
Flyback diodes	[X] Built into the ULN2003A (that's what COM does).
Level shifters	[X] Not needed - hall runs on 3V3, so GPIO34 never sees 5V.
2mm bearing balls	[X] Obsolete since v7.1 - the dot is printed as a dome.

Item	Verdict
Pogo pin connectors	?? Deferred. Do not buy from the CAD window alone. Future contacts need $\geq 1A$ continuous at working compression (prefer 2-3A for power) and a flush measured carrier; the four-wire bus also needs a per-cell 3.3V expander interface.
Crystal / oscillator	[X] ESP32 and the modules have their own.

If you ever DO build the custom muscle board

Not needed now, but for the record - it's `pcb/braillix_muscle_board.kicad_pcb`, 34x44mm:

Part	Package	Note
ATmega328P-AU	TQFP-32, 7x7mm square	0.8mm pitch - not hand-solderable at your level
ULN2003A	SOIC-16 rectangle	Same chip you already own, just SMD
16MHz crystal	HC49	
2x 22pF, 2x 100nF	0402	Tiny - 1.0 x 0.5mm
100 μ F/10V	SMD electrolytic	
10k Ω x2, 4.7k Ω x2	0402	
SS14 Schottky	SMA	?? *this* is the reverse-polarity protection the prototype lacks

Recommendation: order it fully assembled (PCBA) or don't build it. 0402 passives and a TQFP-32 are not a beginner job, and none of it is needed for a working demo.